AETHELRED × M42

Sovereign Verified AI Compute

A verifiable AI infrastructure layer that turns M42's compliance burden into a commercial product, with a quantified path to roughly fifty million dollars of annual value by 2028.

Independently verifiable evidence, data sovereignty, and compute cost discipline for regulated health AI as M42 commercializes its genomics and Med42 assets at national and international scale.

PREPARED BY
Ramesh Tamilselvan, Founder, Aethelred

AUDIENCE
M42 executive and technical review

PURPOSE
Technical and commercial pilot scoping

STATUS
Pre-registration project; Abu Dhabi based; ADGM labs and foundation planned

CONFIDENTIAL SCOPING PROPOSAL · M42 REVIEW MATERIAL

CONTENTS

What this document covers

This proposal sets out a quantified, technical and commercial path for M42 to deploy a verifiable, sovereign AI compute layer across its genomics and clinical AI programs, and to package that capability as a managed service for governments, hospital systems, and pharmaceutical partners. It is written for M42 review only and is structured around a transparent value model that an executive and technical audience can interrogate.

01 · EXECUTIVE SUMMARY

A verifiable compute layer can put roughly $51M of annual value in play for M42 by 2028, validated by a $200K pilot

M42 is commercializing two of the region's most valuable regulated AI assets: the genomic data and pipelines of the Emirati Genome Programme, and the Med42 clinical language models. Selling these to other sovereign governments, hospital systems, and pharma partners raises a question ordinary cloud and internal logs cannot answer alone.

> **Can a buyer, a regulator, or a data-owning government independently verify how a model ran, where it ran, which version ran, and that sensitive data never left its approved boundary?**

Aethelred answers that question and, in doing so, converts a compliance burden into a sellable capability. The value compounds across four streams, sized in Section 8 and summarized here.

~$51M	~$80M	~50%	12×
annual value-at-stake by 2028, across internal savings, managed-service profit, evidence, and premium	cumulative value over the first three years of adoption	gross margin on the managed sovereign verified compute service at scale	illustrative first-year payback on the pilot from a single migrated workload

Three things M42 can do with this layer:

- **Lower the cost of data-intensive compute,** the largest and most immediate value, by routing eligible genomics and AI workloads through decentralized, competitively supplied, verified capacity at a fraction of hyperscaler pricing.
- **Sell trust, not just data and models.** Wrap genomics and Med42 outputs in sovereign, verifiable evidence a foreign government or pharma buyer can audit independently, unlocking premium pricing.
- **Reduce assurance and audit cost** on its own regulated AI by replacing manual log reconstruction with queryable evidence.

THE ASK

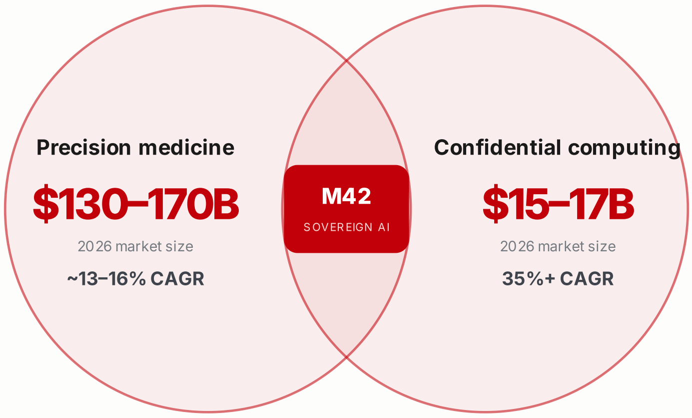
A four-week, pre-testnet sandbox pilot on one non-live workload (a retrospective genomics or Med42 evaluation run), at an indicative cost of about $200K and with no change to M42 production systems. The pilot ends in a measured business case and a clear go / no-go recommendation. A scoping exercise, not a procurement decision.

02 · MARKET CONTEXT

M42 sits at the convergence of two large markets growing at double-digit and triple-digit rates

Precision medicine supplies the demand for genomic and clinical AI; confidential computing is the fastest-growing infrastructure category that makes selling it across borders possible.

EXHIBIT 1 **Two expanding markets, and the position M42 already occupies between them**



M42's proven genomics and Med42 assets sit exactly where accelerating demand meets the verification layer required to sell across borders.

These two markets are precisely where M42 is already selling. As the Emirati Genome Programme and the Med42 models reach governments and pharmaceutical partners abroad, M42 is monetizing genomic and clinical AI at the exact moment demand is accelerating, and at the moment the verification and sovereignty layer required to move sensitive data across borders is becoming one of the fastest-growing categories in infrastructure. The opportunity is therefore one of timing and of capture. The assets are proven; the open question is how to realize their full value internationally, and at what cost.

So what for M42. The largest and most immediate value is on the cost side. Lower compute cost on M42's own genomics and Med42 pipelines, plus a managed service that resells that cost advantage, account for roughly $42M of the ~$51M of annual value at stake quantified in Section 8. The verification layer is what protects and extends that value, and it is credible precisely because it does not depend on trusting M42's own systems.

WHY THIS IS NOT SIMPLY AN INTERNAL BUILD

A team of M42's scale could of course build a verification system. But it would be M42's system, and a foreign government or pharmaceutical buyer would still be asked to trust M42's attestation of its own work. Because Aethelred settles evidence on an independent, decentralized ledger rather than inside a proprietary stack, a counterparty can verify a result directly, without trusting the seller. Independence is the product, and by definition an in-house system cannot supply it. The same openness applies to cost: a decentralized compute market settles pricing transparently, rather than inside an opaque internal allocation.

03 · THE PROBLEM

Five linked constraints leave margin and growth on the table every year M42 waits

Each constraint shares one root cause: there is no standard layer that proves, to an outside party, how an AI workload ran and that its data stayed inside approved boundaries.

EXHIBIT 2 The five constraints and the value each one withholds today

Constraint	Why it limits M42 today	Value withheld
Trust does not travel	An output M42 trusts internally is not independently verifiable when sold to a foreign regulator or partner	Lost premium pricing on genomics and Med42 exports
Sensitive data cannot move	Health and genetic data must stay within approved jurisdictional and consent boundaries	Cross-border deals that cannot clear data-residency review
Audit evidence is weak	Vendor and internal logs are not the same as independently verifiable evidence	High audit-reconstruction cost and slow adverse-event review
Governance is a buying criterion	Buyers increasingly require AI that is explainable, controlled, and reviewable	Contracts lost to providers who can demonstrate assurance
Compute energy is questioned	Large-scale AI draws significant power; conventional blockchain attracts a wasted-energy criticism	Sustainability narrative gap against a public net-zero posture

The cost of the status quo is not zero. It is the sum of margin not captured on exports, deals that stall in compliance, and assurance work done by hand. The value-at-stake model in Section 8 quantifies the upside of closing this gap at roughly $51M per year by 2028.

04 · THE APPROACH

Aethelred makes AI computation provable and settles the evidence, rather than moving tokens

A purpose-built Layer 1 for regulated industries that must demonstrate where a workload ran, which model version ran, and that sensitive data stayed in bounds.

EXHIBIT 3 **How Aethelred differs from general-purpose Layer 1 networks**

Dimension	General-purpose Layer 1 networks	Aethelred
Primary purpose	Generic settlement or smart contracts	Verified AI computation and evidence settlement
Security work	Validators secure the chain without doing useful scientific work	Validators are rewarded for verified, useful AI and scientific work
Enterprise AI fit	Compute, audit logs, and compliance controls sit in separate systems	Compute, attestation, policy, and evidence are bound into one workload flow
Sensitive data	Not designed for genomic or clinical workflows	Built around confidential compute, sovereign policy lanes, and auditable evidence
Evidence model	Transaction history	Digital Seals binding model, workload, policy, attestation, and output commitments

The aim is not to ask M42 to adopt blockchain for its own sake. The aim is to give M42 a verifiable compute and evidence layer that makes its genomics and clinical AI defensible, sovereign, and commercially reusable.

> **PLAIN LANGUAGE**
>
> Think of Aethelred as a notary and a customs office for AI workloads. It does not do the science. It records, provably, that the science was done correctly, in the right place, by the approved model, and it lets an outside party check that record without ever seeing the patient data.

05 · THE PROPOSITION

Value is created across three reinforcing dimensions, bound together by verifiable evidence

Each dimension maps to a stated M42 priority; the connective evidence layer is what turns three internal efficiencies into a capability M42 can sell.

1 Drastically lower compute costs for genomics and AI

This is the value an M42 executive is most likely to act on first, and it is the largest single component of the value at stake. M42 operates some of the most data-intensive pipelines in healthcare: whole-genome sequencing and re-analysis at the scale of the Emirati Genome Programme, Med42 training and evaluation, radiology and pathology inference, and drug-discovery screening. Aethelred decentralizes access to enterprise-grade GPU and TPU infrastructure across multiple approved operators, delivering high-performance raw compute at a fraction of hyperscaler pricing for eligible workloads. **The target for suitable workloads is a 3× to 4× effective improvement, roughly one third to one quarter of comparable cost,** validated workload by workload. Because supply and settlement sit on a transparent, decentralized market rather than an internal allocation, the cost basis is open to inspection rather than taken on trust.

2 Sovereign and compliant data security

The architecture is built with strict data sovereignty as a first principle. Sensitive healthcare and genetic data is processed inside approved confidential-compute lanes, where workload policy metadata fixes jurisdiction, data class, key custody, consent scope, and evidence retention before a job ever runs. Raw records are never exposed to a general operator. This aligns directly with the UAE's national AI ambitions and its data-protection frameworks, and it is precisely the property that allows M42 to process and sell sensitive data across sovereign boundaries with confidence. Because genomic data is sensitive for a lifetime and implicates a patient's relatives and descendants, Aethelred uses post-quantum, quantum-resistant cryptography for its key exchange, signatures, and commitments. This defends against harvest-now, decrypt-later attacks, in which an adversary captures encrypted data today to decrypt once quantum computers mature, and it keeps both the data and the evidence protected for decades rather than only against present-day threats.

3 Sustainability through productive useful work

Aethelred's Proof of Useful Work design ensures that every watt used to secure the network simultaneously performs productive scientific work: inference, genomics pipelines, evaluation runs. This erases the carbon-footprint argument levelled at conventional blockchain, where energy is consumed only to win abstract hash competition. M42 can report a productive-watt ratio and carry a defensible sustainability narrative consistent with its public commitment to health for people and the planet.

The connective thread. Each important workload yields a Digital Seal, a durable, queryable evidence object that lets a regulator or sovereign buyer independently verify the result without exposing patient data. That object is what converts the three dimensions above from internal efficiencies into a premium, sellable service.

06 · THE TECHNOLOGY

Every important workload produces a durable, independently checkable evidence object

The flow runs without changing M42 production systems, beginning with retrospective or synthetic workloads and moving to controlled prospective use only after governance approval.

EXHIBIT 4 **The verified compute and evidence lifecycle, from workload selection to a sealed evidence object**

M42 retains control of the workload, the data boundary, and the acceptance criteria at every step.

6.1 Trusted Execution Environments

A Trusted Execution Environment is a hardware-backed, isolated execution area where the operator of the surrounding infrastructure cannot freely inspect the data or alter the execution path. For M42 this means sensitive health or genomic data can be processed inside a protected boundary, the environment can produce a remote attestation proving what code ran and where, key release can be conditioned on verified enclave identity and approved policy, and audit teams can review evidence without ever touching the raw record.

6.2 Cryptographic verification

Each workload is represented by a signed manifest and a set of cryptographic commitments, so verification does not depend on ordinary logs alone. The signature and commitment schemes are selected to be quantum-resistant, in line with the post-quantum standards now being finalized by NIST, so that a Digital Seal issued today remains both confidential and independently verifiable across the decades-long horizon over which genomic and clinical evidence retains its value.

EXHIBIT 5 **What each cryptographic element proves, and why it matters to M42**

Element	What it proves	Value to M42
Model hash	Which model artifact or version was used	Prevents silent model substitution
Input commitment	The input class or dataset commitment	Review without exposing raw data
Output commitment	The result that was returned	Prevents post hoc result alteration
TEE attestation	Approved code ran in an approved environment	Supports sovereignty and security review
Policy signature	The workload matched approved data, region, and consent rules	Supports governance and regulator review
Digital Seal	The evidence package was finalized and is independently checkable	Creates durable audit evidence

07 · THE TECHNOLOGY

The Digital Seal is what a sovereign buyer's auditors inspect, and what justifies premium pricing

A durable evidence object M42, its customers, or a regulator can review. It proves what happened without disclosing any patient record.

EXHIBIT 6 An illustrative Digital Seal for a genomic interpretation batch

Field	Illustrative value
Seal ID	seal_m42_genomics_2026_000184
Workload	Genomic variant interpretation batch
Model ID	m42_variant_model_v3
Model hash	sha3-256: 8f4a...
Data policy	UAE-resident healthcare data, restricted export
TEE attestation	AMD SEV-SNP or NVIDIA confidential-compute attestation reference
Input commitment	sha3-256 commitment to approved batch
Output commitment	sha3-256 commitment to result set
Proof mode	TEE attestation plus cryptographic commitments; zkML where applicable
Timestamp	2026-06-10T09:42:00Z
Reviewer link	M42 audit or client evidence portal reference

WHY THIS IS THE COMMERCIAL CENTERPIECE

When M42 sells a genomic interpretation to a foreign government, this object is what the buyer's own auditors inspect. It is the difference between asking a customer to trust M42's word and giving the customer a way to verify it. That difference is what justifies premium, sovereign pricing, and it is the mechanism behind the premium uplift in the value model.

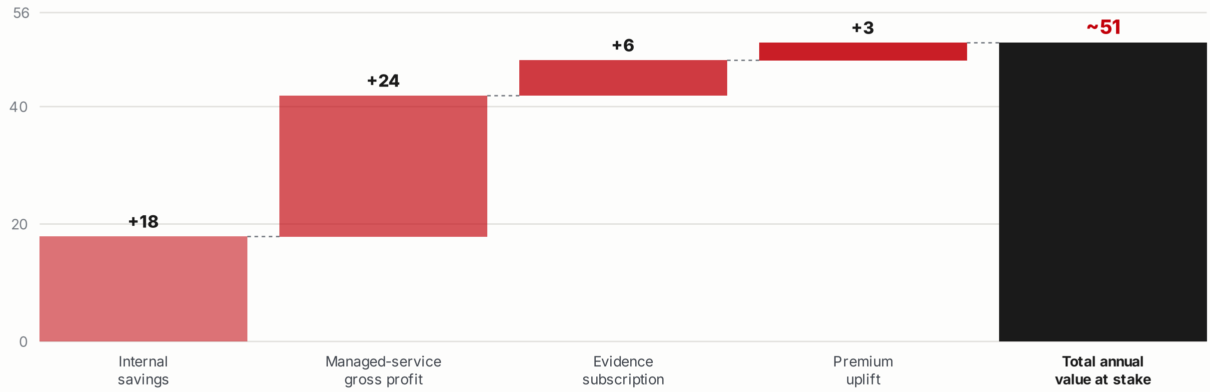
On proof methods, a note for the technical reviewer. zkML verification is used for suitable inference workloads where proof generation is tractable today. Larger or early pilot workloads rely on attestation-backed evidence and cryptographic commitments while zkML coverage is expanded. The pilot documents which mode applies, so the assurance level is explicit rather than implied.

08 · VALUE AT STAKE

Four reinforcing streams build to roughly $51M of annual value by 2028

The bridge below decomposes the steady-state annual value into its sources. Figures are illustrative planning scenarios built on the assumptions footnoted below and validated through the pilot.

EXHIBIT 7 Annual value-at-stake bridge for M42 at steady state, 2028 (US$M)



$17.9M Internal compute savings on eligible genomics and AI workloads	**$24.0M** Managed-service gross profit from third-party compute displaced	**$6.0M** Evidence and assurance subscription revenue	**$3.0M** Premium uplift on existing M42 services from verified differentiation

Assumptions (illustrative, to be validated by pilot). Addressable M42 AI and genomics compute spend of about $40M in 2026 growing roughly 22% per year; effective cost reduction of 60% on eligible workloads; eligibility ramping to about 50% of spend by year three; third-party client compute displaced of about $80M by year three at a 30% M42 gross-profit margin on client baseline; evidence subscription and premium uplift as shown. All figures depend on workload mix, compute supply, and adoption, and apply to eligible workloads rather than M42's entire footprint.

09 · COMMERCIAL CASE A

Internal savings alone could fund the entire build, even under conservative assumptions

Annual savings equal eligible compute spend multiplied by the validated reduction rate. The matrix below shows the range; even the conservative corner is material.

EXHIBIT 8 Sensitivity: illustrative annual internal savings by eligible spend and cost reduction (US$M)

nd ↓ / Reduction →	50% reduction	60% reduction	70% reduction
$20M	$10M	$12M	$14M
$40M	$20M	$24M	$28M
$60M	$30M	$36M	$42M

Annual internal savings = eligible spend × effective reduction. Base case (~$40M eligible, 60% reduction) highlighted in the model at ~$24M.

The pilot should not attempt to prove every workload at once. The best first workloads are repeatable, measurable, and non-live clinical.

EXHIBIT 9 Candidate first workloads, ranked for clean measurement

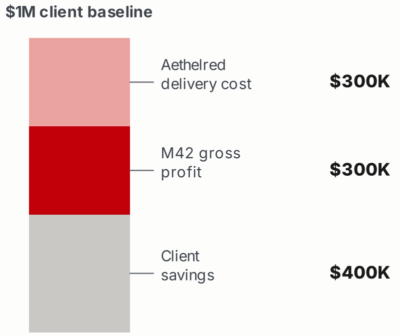
Candidate workload	Why it is a good first pilot	Primary metric
Genomic variant calling or interpretation	High volume, repeatable, clear baseline comparison	Cost per run, runtime per sample batch
Med42 evaluation or fine tuning	Strategic AI asset with measurable benchmark outputs	Cost per evaluation or fine tuning job
Retrospective radiology AI evaluation	Clinically relevant, can start on historical data	Cost per study, evidence completeness
Drug discovery screening	Compute heavy and commercially valuable	Cost per candidate screen or batch

10 · COMMERCIAL CASE B

The managed service carries roughly a 50% gross margin while saving clients 35–45%

M42 packages the capability as M42 Sovereign Verified AI Compute, powered by Aethelred, for governments, hospital networks, and pharma. Both sides win because Aethelred lowers the underlying delivery cost.

EXHIBIT 10 Unit economics per $1M of client compute spend



M42 price to client: **$600K** · gross margin **50%**
Client saves **$400K** (40%) versus hyperscaler-equivalent.

EXHIBIT 11 Portfolio economics by client compute displaced (US$M)

Displaced	M42 revenue	M42 profit	Client save
$10M	$6M	$3M	$4M
$50M	$30M	$15M	$20M
$100M	$60M	$30M	$40M

M42 revenue at 60% of client baseline, delivery at 30%, gross profit 30%. Illustrative; actual economics depend on adoption volume, workload eligibility, and assurance level.

Buyers include government health programs, national AI programs, hospital and diagnostic networks, biopharma, clinical research organizations, and regulated enterprises that need AI compute but cannot accept black-box infrastructure.

The commercial logic is simple. Clients save materially against current spend while M42 keeps a managed-service margin, because Aethelred reduces the underlying delivery cost and supplies the evidence layer that makes the service sellable into regulated, sovereign buyers.

11 · ADOPTION TRAJECTORY

Value compounds from about $4M in year one to roughly $51M by year three

Internal savings come first and de-risk the path; managed-service revenue and evidence subscriptions layer on as external clients adopt. Cumulative value over three years is about $80M.

EXHIBIT 12 Annual value-at-stake by stream across the three-year ramp (US$M)

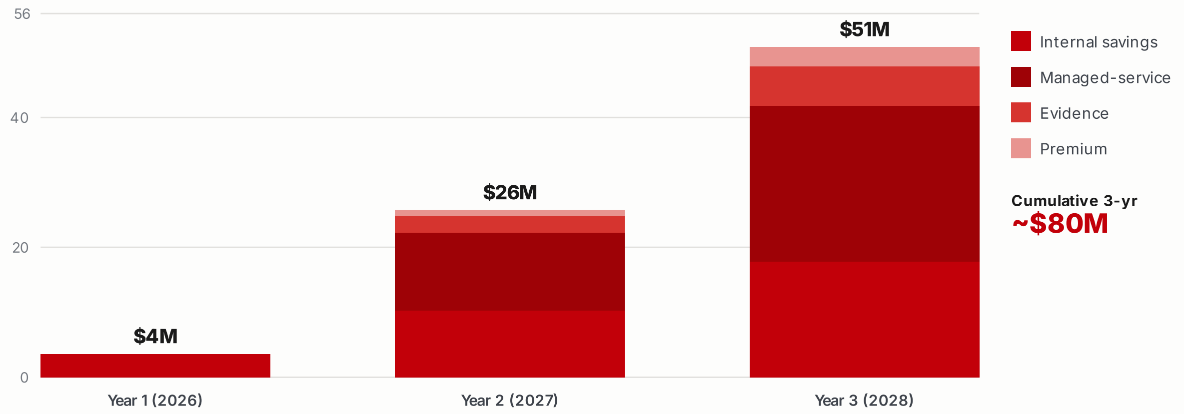


EXHIBIT 13 What happens in each phase of the ramp

Phase	Focus	Outcome for M42
Year 1	Internal workloads migrate	Cost savings on eligible genomics and Med42 pipelines; evidence model proven on real workloads
Year 2	First external clients	Managed service launches with a small number of sovereign and enterprise buyers; assurance subscriptions begin
Year 3	Sovereign platform at scale	Managed service and evidence revenue dominate; M42 owns the trusted compute and evidence layer for national health AI

AN OPTIONAL DEEPER ROLE: M42 AS A VALIDATOR, NOT ONLY A CLIENT

Because Aethelred secures the network through Proof of Useful Work, the work that secures the chain is itself productive AI and genomics computation. That lets M42 take a second role over time. Alongside being the anchor client, M42 could operate validator capacity on its approved infrastructure, contribute useful compute, and share in network rewards, turning a procurement line into a stake in the platform it helps anchor. Two conditions preserve the value. First, the independence that makes the evidence sellable requires that no single party, including M42, can control consensus, so M42 would be one validator within a deliberately diverse set, while correctness stays enforced by hardware attestation and cryptography rather than by validator identity, and confidential compute ensures a validator never sees another party's raw data. Second, the treatment of network rewards and any token held in treasury would be structured and reviewed under the ADGM virtual-asset framework, with tax and accounting input. This is a phase-two option to explore once the pilot has proven the core, not a precondition for it.

12 · DILIGENCE

The opportunity is robust because M42 takes no irreversible step during the pilot

A serious reviewer will test this against its weakest points. Each principal risk is stated plainly with the mechanism that contains it.

EXHIBIT 14 **Principal risks and how the pilot contains each one**

Risk	The concern	How it is managed
Project maturity	Aethelred is pre-registration with no live public testnet	The pilot is a contained sandbox on non-live data with a go or no go gate; no production dependency is created
Vendor dependency	Reliance on a single early stage provider	Aethelred supplies a thin coordination and evidence layer over M42 approved compute; the Seal format and evidence are portable and documented
Data security	Exposure of patient or genomic data	Confidential compute and TEE boundaries; the pilot uses retrospective or synthetic data; raw records never leave the approved environment
Regulatory	Cross border and clinical compliance	Policy metadata encodes jurisdiction, consent, and export rules; alignment to UAE data protection, NIST AI RMF, FDA GMLP, and ISO and IEC 42001
Performance and assurance	zkML may not cover all workloads; cost claims unproven	Proof mode is documented per workload; cost is measured against M42 baselines and treated as a hypothesis to validate
Commercial	Unclear path from pilot to value	The pilot ends in a quantified business case with explicit internal savings, client revenue, and assurance metrics

GOVERNANCE AND HONESTY OF STATUS

Aethelred is a pre-registration project led by Ramesh Tamilselvan, based in Abu Dhabi, with Aethelred Labs and the Aethelred Foundation planned for ADGM registration. Commercial commitments, procurement, regulated activity, and public communication should occur only after appropriate entity setup, legal review, and approvals. This document is a scoping proposal, not a procurement decision or a claim of an existing relationship with M42.

13 · THE PILOT

A $200K, four-week pilot is the lowest-risk way to validate a multi-year opportunity

One workload, four phases, ending in a measured business case and a go / no-go decision. The asymmetry is the point: a small, contained cost against a path to roughly $51M of annual value.

EXHIBIT 15 Proposed four-week pilot structure

Phase	Duration	Output
Workload selection	1 week	Choose one genomics, Med42, or clinical AI workload
Baseline measurement	1 to 2 weeks	Current cost, runtime, accuracy, throughput, and evidence process
Aethelred sandbox run	2 to 4 weeks	Cost comparison, evidence objects, security model, operational notes
Business case report	1 week	Savings, revenue, sustainability, and a go or no go recommendation

EXHIBIT 16 The pilot asymmetry

Pilot cost

~$0.2M

one-time, four weeks

Validates a path to

~$51M / yr

annual value-at-stake by 2028

A single migrated workload returns
~12× in year one alone

RECOMMENDED FIRST WORKLOAD

A retrospective, non-live genomics interpretation or Med42 evaluation run. This avoids any patient-care risk, creates a clean measurable baseline, and maps directly to the assets M42 is already commercializing internationally.

THE NEXT STEP

A technical and commercial workshop to lock the pilot scope

A working session with M42 stakeholders to agree the first workload, the current baseline cost and infrastructure, the acceptable data policy and security boundary, the required evidence outputs, the success metrics for internal savings and client revenue, and the path from pre-testnet sandbox to testnet pilot.

EXHIBIT 17 **Suggested workshop participants**

Side	Participants
M42	AI infrastructure, genomics or life sciences, clinical AI, information security, business development
Aethelred	Ramesh Tamilselvan, with technical architecture and pilot implementation support

The goal is deliberately narrow. Prove one high-value M42 workload, measure it honestly against the value model in this document, then expand only if the numbers and the evidence are strong.

14 · APPENDIX

References, model assumptions, and important notice

Market and M42 sources

- Precision medicine market sized at roughly $125 to $172 billion in 2026, growing at a low-to-mid teens CAGR (multiple 2026 industry analyses, including Mordor Intelligence and Precedence Research)
- Genomics market sized at roughly $38 billion in 2026, growing at about a 12.7% CAGR (Fortune Business Insights, 2026)
- Confidential computing market sized at roughly $15 to $17 billion in 2026, the fastest-growing infrastructure category, with reported CAGRs from 35% to above 60% (Mordor Intelligence, Fortune Business Insights, The Business Research Company, 2026)
- M42 corporate scale, Emirati Genome Programme, and Med42 model details from M42 public reporting and publications, m42.ae

Standards and regulatory anchors

- UAE data protection framework, u.ae digital UAE data pages
- NIST AI Risk Management Framework
- FDA Good Machine Learning Practice for Medical Device Development, guiding principles
- ISO and IEC 42001:2023, Artificial Intelligence Management System
- NIST post-quantum cryptography standards, FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA)

Live hyperlinks for the standards above should be inserted in the final distributed file and link checked before sending.

Model assumptions

The value-at-stake figures are illustrative planning scenarios, not forecasts. They assume an addressable M42 AI and genomics compute spend of about $40M in 2026 growing roughly 22% per year, an effective cost reduction of about 60% on eligible workloads, an eligibility ramp to about half of spend by year three, third-party client compute displaced of about $80M by year three at a 30% M42 gross-profit margin on client baseline, and evidence and premium revenue as shown in Section 8. The pilot exists precisely to replace these assumptions with measured figures.

IMPORTANT NOTICE

This document is a confidential business case and technical scoping proposal prepared for M42 review. It is not an investment pitch, legal or regulatory advice, or a claim of an existing commercial relationship with M42. All financial figures are illustrative planning scenarios to be validated through a controlled pilot before any commercial reliance. Aethelred is a pre-registration project; commercial commitments should follow appropriate entity setup, legal review, and approvals.

AETHELRED · VERIFIABLE AI INFRASTRUCTURE

Prepared by Ramesh Tamilselvan · Abu Dhabi · June 2026